

A Self-Closing Gate Conundrum

#1 Victory Royale

Element F: Consideration of Design Viability

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

Our solution will be realistic to construct, because our group will buy the main components of our design and assemble them together by putting the end of the spring through the galvanized hanger straps hole, then put cut cloth around the galvanized hanger strap then we are done. The spring that we will use for our prototype is a tension spring which when compressed closes the gate. When the spring is compressed it wants to return to its original state which is when the spring has no force acting upon itself.

A few features of our design that go beyond our design requirements are how far our tension can bend before breaking. The strength needed to break the clamp that is connected to the gate and the spring.